Mating inbreeding and hetrosis

Systems of Mating

- Mating is one of the two breeding tools (i.e. selection and mating) used by breeders to make genetic change

- Mating determines which (selected) males are bred to which (selected) females

- It is the making of pairs among the parents that are already selected

- There are many different methods for mating: – based on expected animal performance:

  - Simple random mating; Positive assortative mating; Negative assortative mating

Systems of Mating…

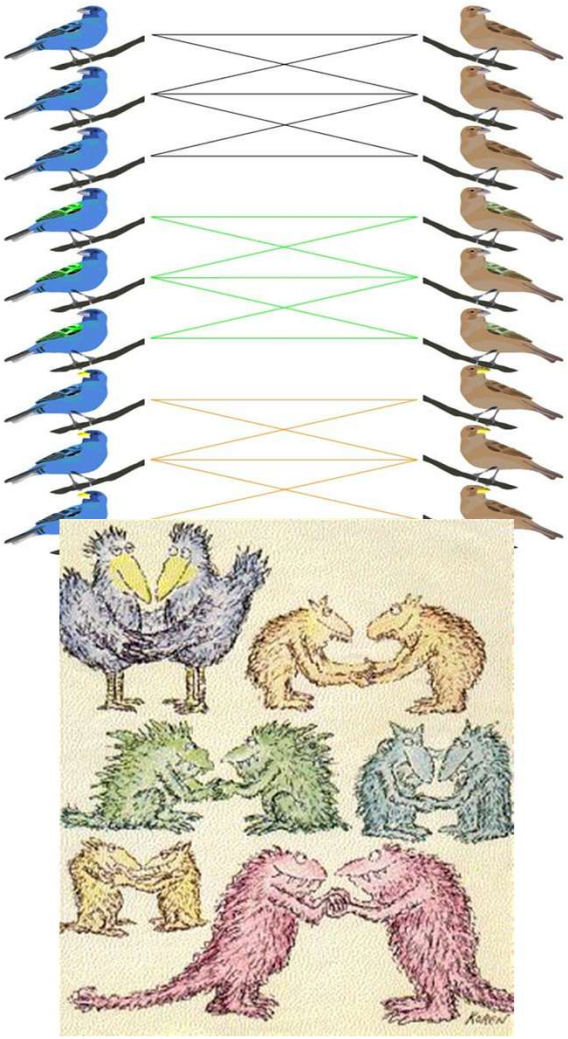
- Methods for mating based on pedigree relationship
- Pure breeding systems
  - Inbreeding
  - Line breeding
  - Outbreeding
- Crossbreeding systems
  - Discontinuous
  - Continuous crossbreeding

Systems of Mating Cont . . .

- **Random Mating**
- Mates are chosen at random
- Any individual has an equal chance of mating with any other individual of the opposite sex in the selected population of individuals
- No systematic mating scheme
  - It is easy and may not require performance records or genetic predictions
  - Used in genetic evaluation of bulls in dairy cattle to avoid bias
    - e.g.: a sire is assigned to a sufficiently large number of mates which are chosen at random:
      - Its evaluation neither benefits from having a particularly good set of mates nor suffers from a particularly poor set

Positive Assortative Mating

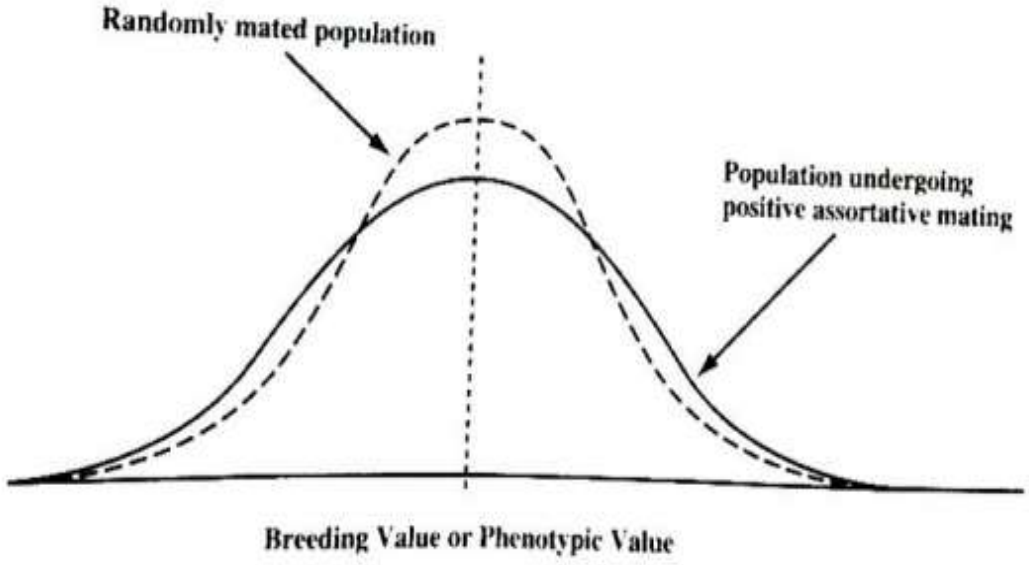
- Involves mating of similar individuals, such as:
  - Mating males with the highest performance to females with highest performance
  - The tallest males to the tallest females
  - Biggest to the biggest
  - Smallest to the smallest
  - Fastest to the fastest, etc.
  - Usually used to produce ***extreme*** **and** ***truly superior*** **individuals with the aim of using them to impact the next generation**



Positive Assortative Mating . . .

- It tends to create more genetic and phenotypic variation in the offspring generation compared to randomly mated population, i.e.,

– Mating of the highest to the highest and lowest to the lowest spreads the distribution away from the center



Negative Assortative Mating

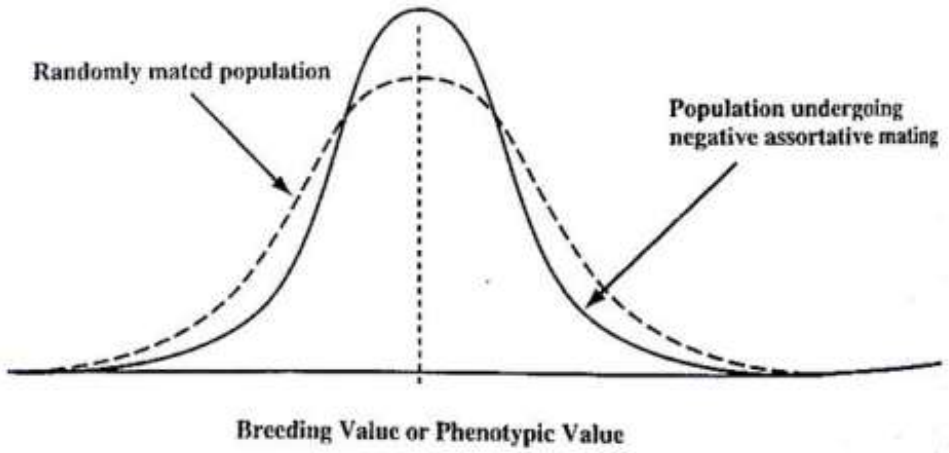
- Involves mating of dissimilar individuals, such as:
- The tallest males to the shortest females – Biggest to the smallest – Fastest to the slowest, etc.
- It is not a good strategy when the chief aim is to speed the rate of directional genetic change as it reduces genetic variation

- However, it enables to increase and maintain phenotypic uniformity at a desired intermedium optimum

Negative Assortative Mating . . .

- It tends to **reduce genetic and phenotypic variation** in the offspring generation compared to randomly mated population, i.e.,

– Mating of animals that are extreme in the opposite direction tends to produce more intermediate types and reduces the number of extreme offspring

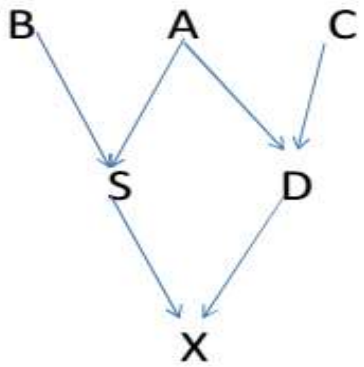


Pure breeding Systems:

Inbreeding

- This is the mating of relatives – individuals more closely related than the average for the population
- It was practiced at high levels in the establishment of breeds

e.g.



- X is inbred because its sire (S) and dam (D) are half sibs, the parents have the same sire (A)
- In other words, A is a common ancestor to the parents of X

Cont…

I. Close inbreeding:

-denotes mating of closely related individuals like ***Dam to Son*** or ***Sire to Daughter*** or ***Full Brother to Full Sister***.

-is breeding of individuals having r/nship above 0.25. The different types of progeny generated are monozygotic twins (clones), full-sibs & haff-sibs having genetic r/nship as 1.0, 0.5 & 0.25.

- **Selfing**
- **Full-sib mating**
- **Half-sib mating**

Cont…

II. Mild inbreeding:

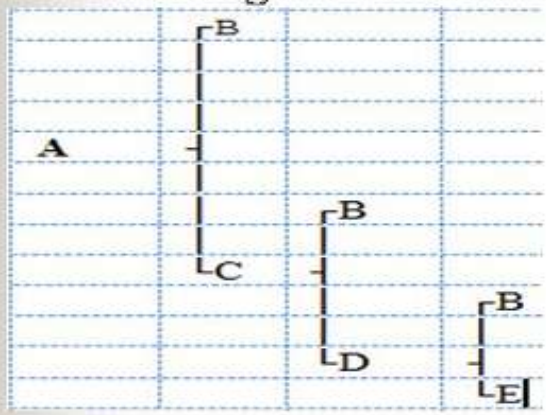
- denotes mating of sibs beyond 2nd generation up to 4th generation i.e. mating of sibs from 3rd generation & 4th generation.

In this case the inbreeding is milder than close-inbreeding i.e. full-sib or half-sib mating

Cont…

Line breeding

- is the practice of mating individuals that are descendents of one particular outstanding ancestor. An attempt should be made to keep inbreeding as low as possible when starting a line breeding program.



- The diagram to the left is a good example of **line-breeding**.
- From the above diagram we see that:
  - "**A**" has 87.5% of **B´s** genes;
  - "**C**" has 75% &
  - "**D**" has 50%.

The different alphabets represent different names of the same breed.

Possible use of inbreeding

- To determine the **actual genetic worth** of an individual
  - Mate a sire to 23 to 35 of his own daughters
  - **Aim:–**
    - *Test for all recessive genes the sire may carry*
    - *Find out an indication of the **desirable genes** he may possess*
    - *To **eliminate** faulty recessive alleles from a population*
    - *To increase phenotypic uniformity*

      • This is usually for qualitative traits such as coat color, horn, etc. – To develop lines that can be used for crossing, • e.g. poultry, swine, and hybrid seed corn

Effects of inbreeding

- **Increased homozygosity (uniformity)**
  - It increases the number of homozygous loci in inbred individuals
  - Increases the frequency of homozygous genotypes in an inbred population
- **Expression of deleterious recessive alleles with major effects**
  - An aspect which gives inbreeding a bad reputation

***i.e* Inbreeding depression**

- Reduction in the mean performance (phenotypic value) of inbred for reproductive and fitness or survivability traits
- Inbreeding depression is the reverse of hybrid vigor

Coefficient of Inbreeding (F)

- The coefficient of inbreeding (F) is the probability that two alleles at a randomly chosen locus are identical by descent (IBD) result of the mating of related individuals in the population since the reference date
- IBD = *copies of same alleles from common ancestor*
- F ranges from 0 to 1
- The coefficient of inbreeding for animal X is commonly symbolized as F x (called, "F of x")

Calculation of Inbreeding & Relationship Coefficients

A. Calculation of Inbreeding Coefficient

The Inbreeding Formula

The formula for the inbreeding coefficient of animal X is:

$$F_X = \sum_{CA=1}^{k} \left(\frac{1}{2}\right)^{n_1+n_2+1}(1 + F_{CA})$$

where CA = a common ancestor of the sire and dam of X
k = the number of common ancestors in X's pedigree
n_1 = the number of generations separating the common ancestor from the sire of X
n_2 = the number of generations separating the common ancestor from the dam of X
F_{CA} = the inbreeding coefficient of the common ancestor

B. Calculation of Relationship Coefficient

The Formula for Wright's Coefficient of Relationship

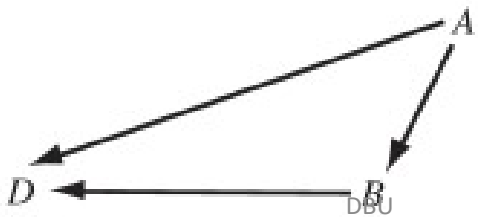
The formula for Wright's coefficient of relationship between individuals X and Y is:

$$R_{XY} = \frac{\sum_{CA=1}^{k} \left(\frac{1}{2}\right)^{n_1+n_2}(1 + F_{CA})}{\sqrt{1 + F_X}\sqrt{1 + F_Y}}$$

- In the relationship formula, $\mathbf{n_1}$ is the number of generations separating the common ancestor from *X*, not *X*'s parents and $\mathbf{n_2}$ is the number of generations separating the common ancestor from *Y*

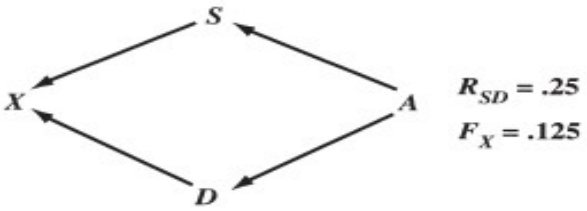
Rules of path method

- Each pair of paths must start from the common ancestor and, always following the direction of the arrows in the diagram
- To avoid double counting, not include the same individual (with the exception of the common ancestor) “no bottleneck” rule
- In certain situations an animal is considered to be its own common ancestor (separated from itself by zero generations)

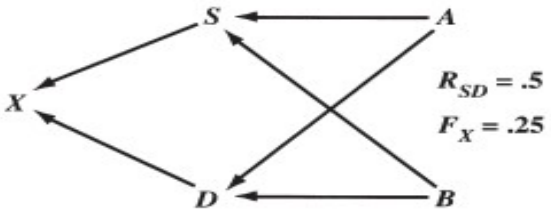


Inbreeding and Relationship coefficients of relationship in common mattings

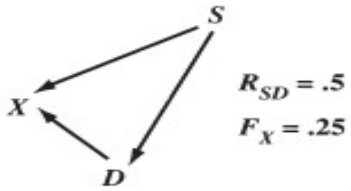
Half-sib mating



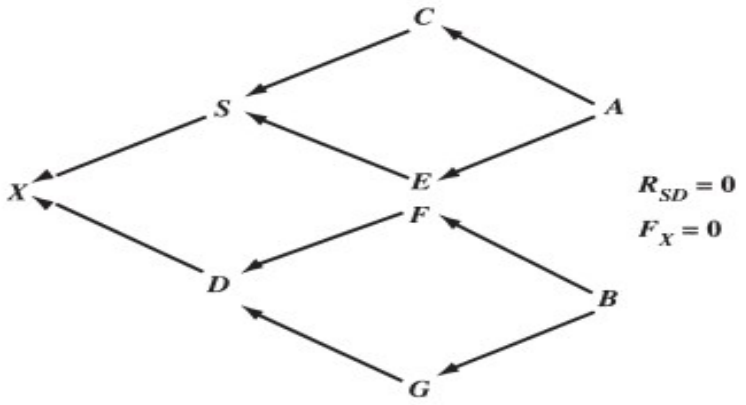
Full-sib mating



Sire × daughter mating



Mating of unrelated inbreds



Calculating Inbreeding and Relationship Coefficients Using the Path Method

Steps:

1. Convert the pedigree to an arrow diagram in which each individual appears only once

2. Identify all possible CAs, identify every unique pair of paths by which identical genes can be transmitted from the common ancestor to the sire and dam of X

3. Identify those CAs which are inbred

4. Fill in the following table:

Common Ancestor	Paths	$(\frac{1}{2})^{n_1+n_2+1}$ for F_X or $(\frac{1}{2})^{n_1+n_2}$ for R_{XY}	$1 + F_{CA}$	Product of Last Two Columns

Exponential Values of □

$(½)^2 = 0.250000$	$(½)^6 = 0.015625$
$(½)^3 = 0.125000$	$(½)^7 = 0.007112$
$(½)^4 = 0062500$	$(½)^1 = 0.003906$
$(½)^5 = 0.031250$	$(½)^9 = 0.001953$

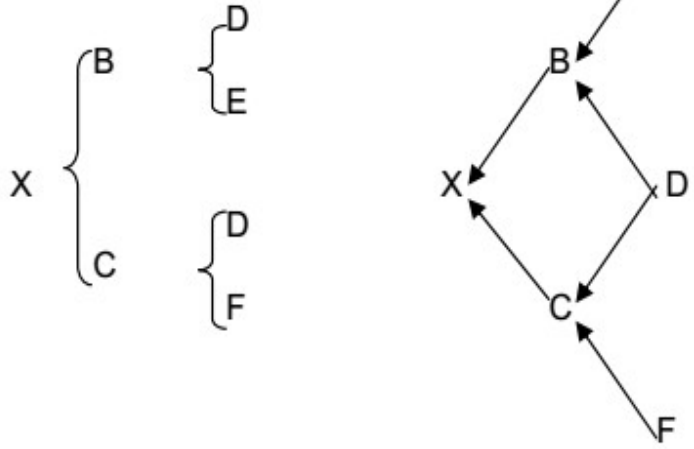
0.0625

- Sum the last column to compute either F_X or the numerator of R_{XY}

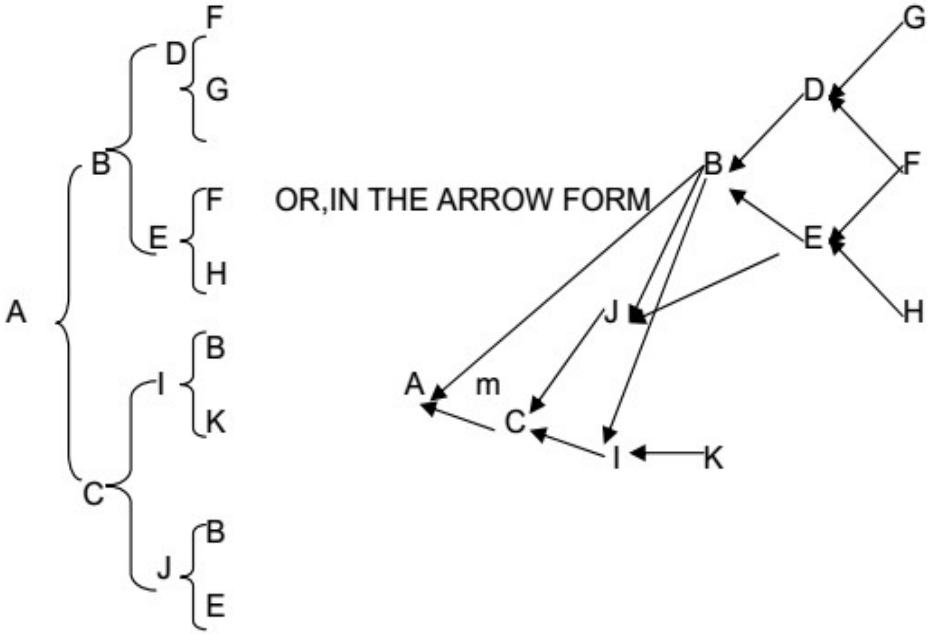
6. Divide the sum by $\sqrt{1 + F_X}\sqrt{1 + F_Y}$. to find RXY

** If CAs are not inbred, then $1+F_{CA} = 1$*

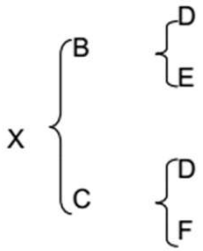
(A) Bracket and arrow-style pedigree of an animal (X) produced by mating of half-brother and half sister

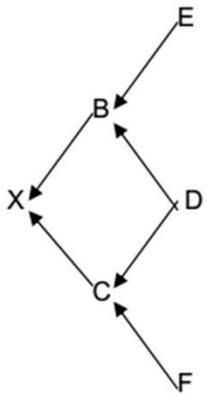


(B) Bracket and arrow-style pedigree of an inbred common ancestor



Calculation of Inbreeding and relationship in the above mating (A)





CA	Path	n	$(\frac{1}{2})^{n+1}$	$1+F_{CA}$	$(\frac{1}{2})^{n+1}(1+F_{CA})$
D	B & C	2	$(\frac{1}{2})^3 = 0.125$	1+0	0.125 X 1=0.125
					Total = **0.125**

Therefore, inbreeding coefficient of inbreeding of inbred animal "X" F_X =0.125 or 12.5%

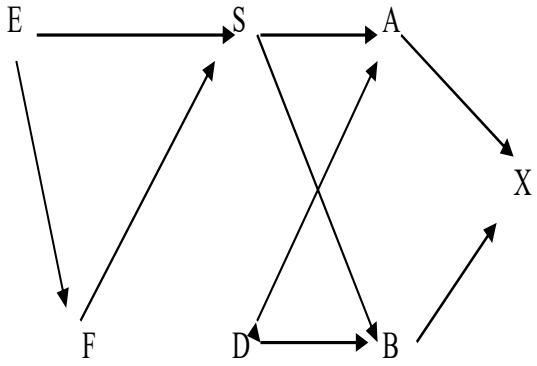
CA	Path	n	$(\frac{1}{2})^{n}$	$1+F_{CA}$	$(\frac{1}{2})^{n}(1+F_{CA})$
D	B & C	2	$(\frac{1}{2})^2 = 0.25$	1+0	0.25 X 1=0.25
					Total = **0.25**

$$0.25/\sqrt{1+F_B} .\sqrt{1+F_C}$$

$$0.25/\sqrt{1+0} .\sqrt{1+0} = 0.25$$

Therefore, relationship coefficient of animal B & C =0.25 or 25%

Calculation of Inbreeding of Individual X



CA	Path	n	$(\frac{1}{2})^{n+1}$	$1+F_{CA}$	$((\frac{1}{2})^{n+1})(1+F_{CA})$
S	A& B	2	$(\frac{1}{2})^3 = 0.125$	1+0.25	0.125 X 1.25=0.1562
D	A&B	2	$(\frac{1}{2})^3 = 0.125$	1+0	0.125X1= 0.125
					Total = **0.2812**

* Therefore, inbreeding coefficient of inbreeding of inbred animal "X" FX = 0.2812

What does the coefficients tell to a breeder ?

The IC measures the % increase in homozygosity in an individual relative to the average of the breed.

If an individual has an IC of 0.25, it means that he is expected to have 25% more homozygous gene pairs than a non-inbred individual from the same population.

The IC can have any value b/n 0 & 1 although it isunlikely for it to have a value much above 0.5 in most herds of livestock. It is fairly easy to have it approach 1 in plant species where self-fertilization is possible.

There are some lines of laboratory animals with very high average ICs but since it takes several generations of full-sib or parent-offspring matings to have an IC over 0.5 it is unlikely that many farm animals will have extremely high

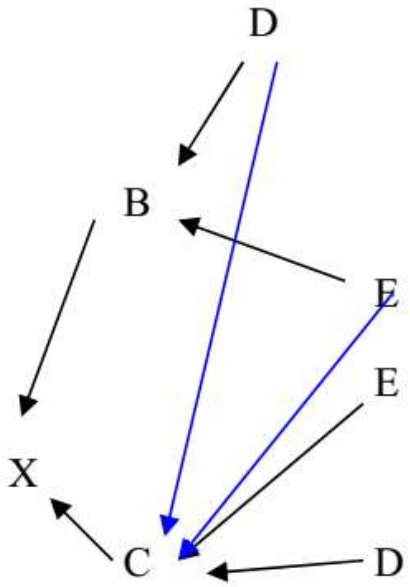
What does the coefficients. . . ?

R measures the probability that 2 individuals have a particular gene in common b/c of common ancestry.

The r/nship b/n a parent & offspring or b/n fullsibs is 0.5 while the r/nship b/n 2 half-sibs or an individual & its grandparent is 0.25.

EXERCISE: 1

1. Calculate the IC of individual “X”

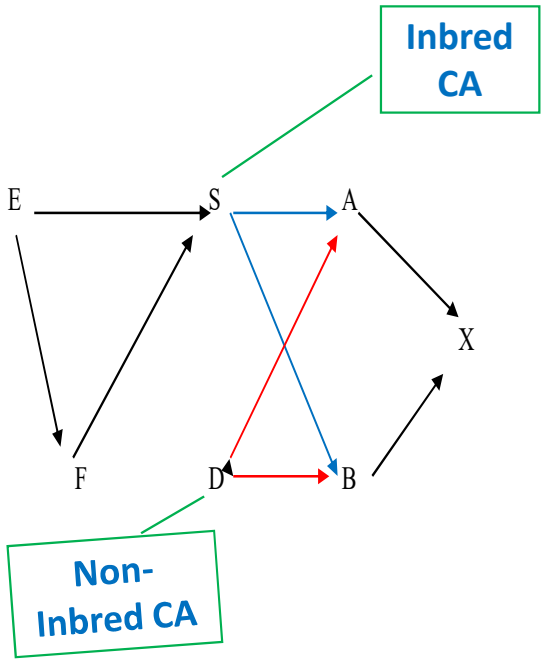


CA	Path	n	$(\frac{1}{2})^{n+1}$	$1+F_A$	$((\frac{1}{2})^{n+1})(1+F_A)$
D	B &C	2	$(\frac{1}{2})^3 = 0.125$	1+0	0.125 X 1=0.125
E	B &C	2	$(\frac{1}{2})^3 = 0.125$	1+0	0.125 X 1=0.125
					Total = **0.25**

* Therefore, inbreeding coefficient of inbreeding of inbred animal “X” FX = 0.25 or 25%

EXERCISE . . .

2. Calculate the IC of individual “X”

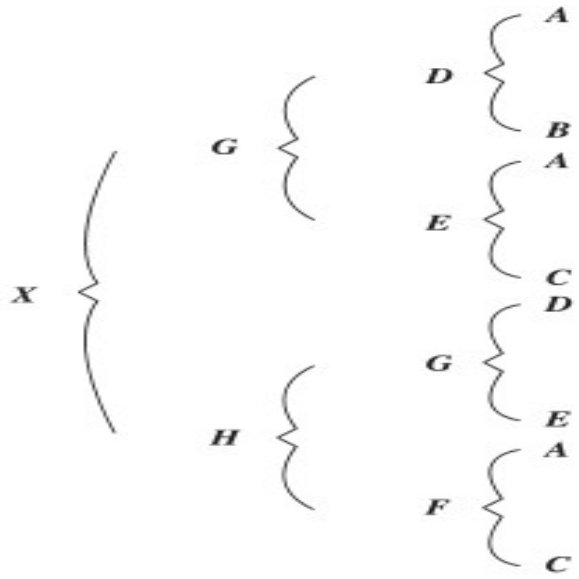


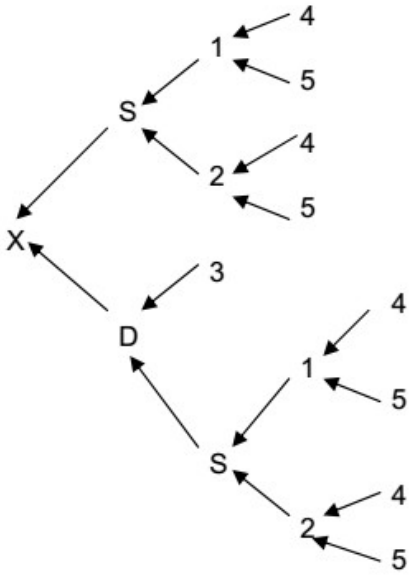
CA	Path	n	$(\frac{1}{2})^{n+1}$	$1+F_A$	$((\frac{1}{2})^{n+1})(1+F_A)$
S	A& B	2	$(\frac{1}{2})^3 = 0.125$	1+0.25	0.125 X 1.25=0.15625
D	A&B	2	$(\frac{1}{2})^3 = 0.125$	1+0	0.125X1= 0.125
					Total = **0.2812**

* Therefore, inbreeding coefficient of inbreeding of inbred animal “X” FX = 0.2812

EXERCISE . . .

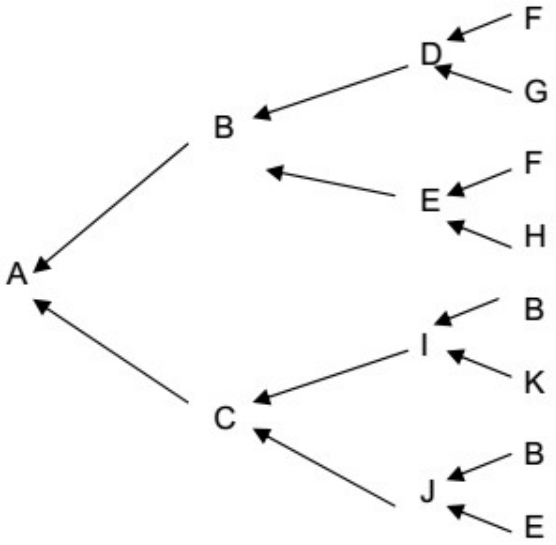
3. From the pedigrees given below, calculate the inbreeding and relationship coefficient of the individual X (Fx) and also the type of mating i.e. full sib , half sib, parent offspring mating etc involved in producing the individual X





EXERCISE . . .

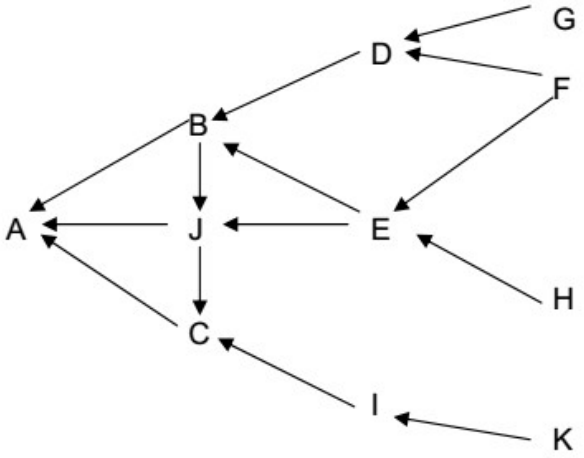
4. Calculate the inbreeding coefficient of individual ‘A’, ‘B’ and ‘C’ from the given pedigree



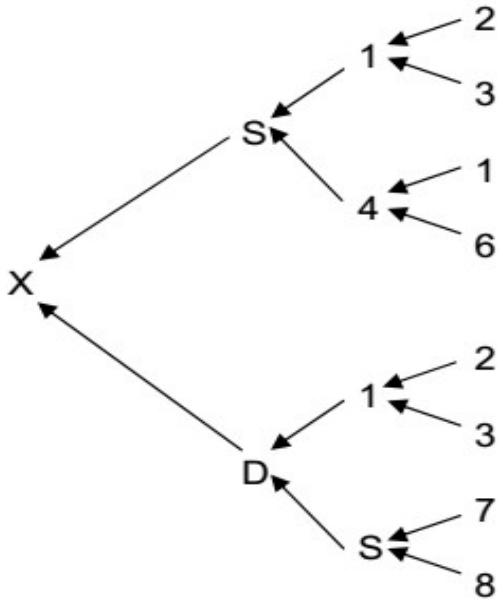
EXERCISE . . .

5. Calculate the relationship coefficient between:

a) A & B



b) S & D



Inbreeding rate

Effective/Active population size (Ne) & Inbreeding rate (ΔF)

- **Inbreeding occurs due to the mating of relatives**

- **In a closed population inbreeding is inevitable**

✓**Inbreeding rate describes the increase in F over time**

Inbreeding rate cont. . .

❖**Inbreeding rate (ΔF) is determined by the *effective/active population size (Ne) rather than the actual/total population size***

❖**Consider which of these two ppn´s would have the higher inbreeding rate (ΔF)**

- **499 breeding females & 1 breeding male ?**
- **250 breeding females & 250 breeding males ?**

Calculating N_e

$$Ne = \frac{4.N_m.N_f}{N_m + N_f}$$

Accounting for unequal sex ratio

- **Effective ppn size** (Ne) reduces towards sex with fewer breeding individuals

Calculating inbreeding rate (ΔF) from Ne

- Larger Ne→ ……→ΔF decreases
- Smaller Ne→ …… →ΔF increases

ΔF & Ne are inversely related by the formula:

$$\Delta F = \frac{1}{2Ne}$$

The rate of inbreeding

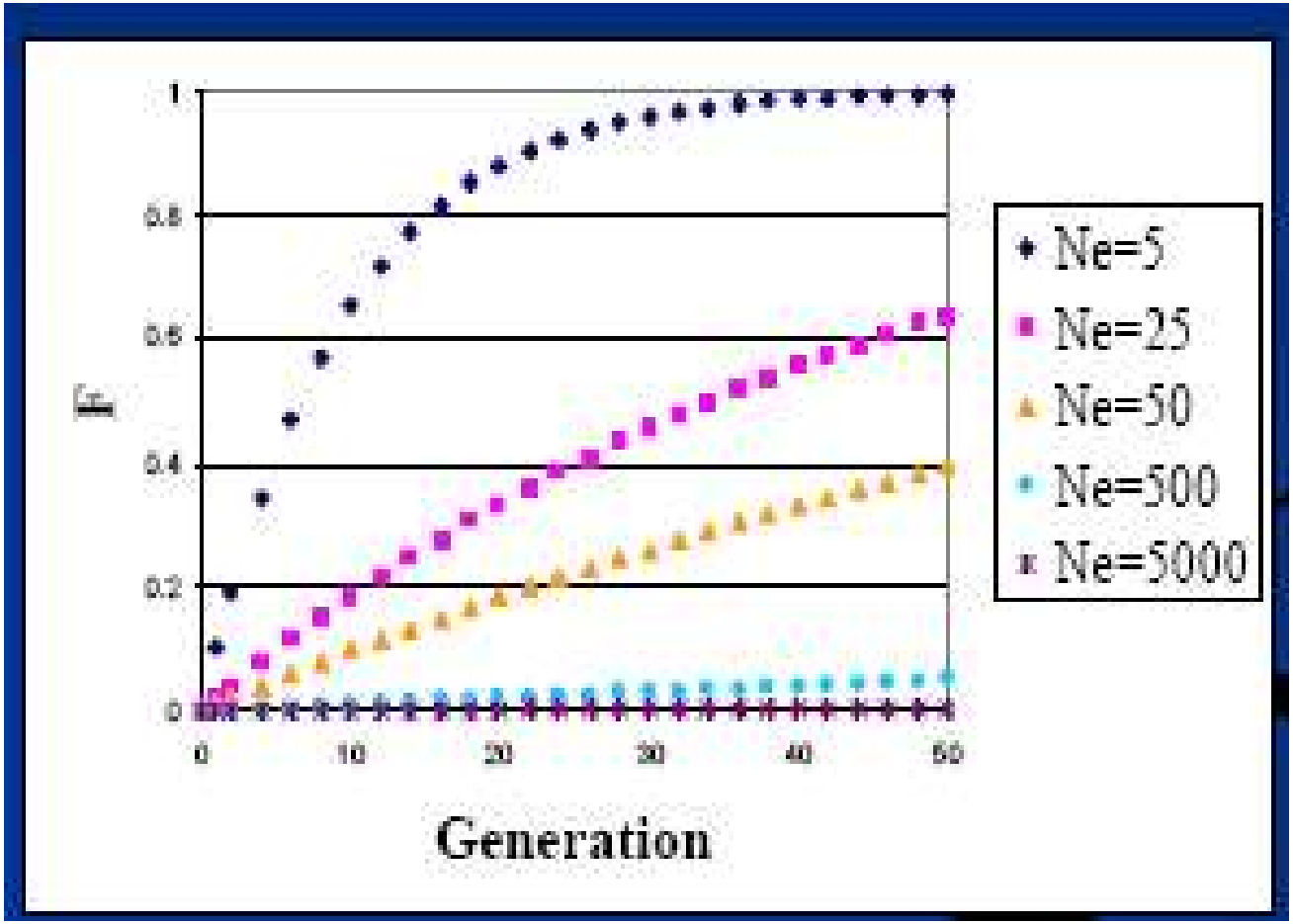
- The rate of inbreeding (F_t) at time 't' can be calculated as:

$$F_t = 1 - \left[1 - \frac{1}{2Ne}\right]^t$$

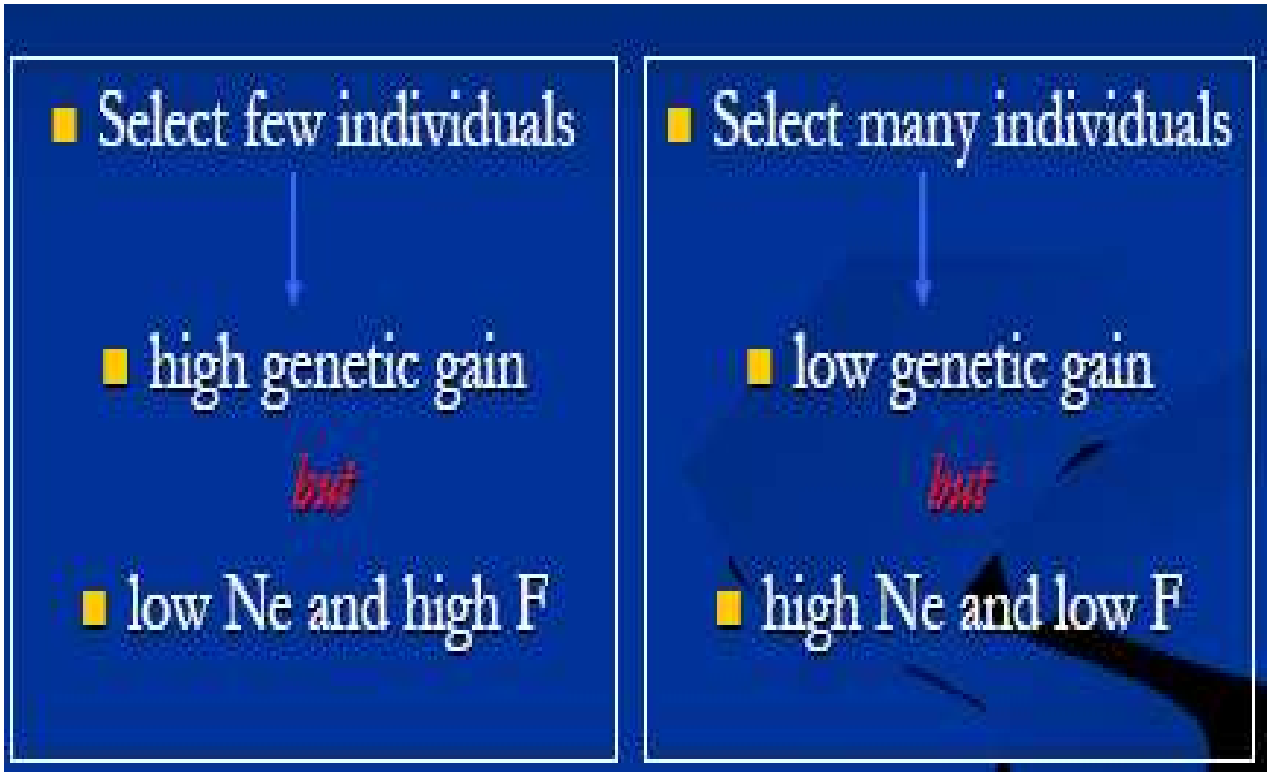
where t is number of generations

* Note that this only holds in the absence of *selection* & *random mating*

The ΔF & Ne


1
0.8
0.6
0.4
0.2
0
0
10
20
30
40
50
Generation
Ne=5
Ne=25
Ne=50
Ne=500
Ne=5000

Balancing genetic gain & inbreeding



Need to balance rates of F & genetic gain

Predicting the rate of inbreeding

- Provides an indication of the expected increase in inbreeding depression, it would be nice to be able to predict the rate of inbreeding in future generations
- Predicting the exact future is not possible, but you can make an approximation
- If you know **the number of males and females** that are used for breeding, you

can predict what the rate of inbreeding will be

- Simple formula as below can be used

$$\Delta F = \frac{1}{8N_m} + \frac{1}{8N_f}$$

* *But the exact rate of inbreeding will depend on the genetic relationships between the animals, and that is not taken into account in this formula and* the rate of inbreeding depends on your population size, more than on mating strategy

Outbreeding

- It is the opposite of inbreeding
- **Outbreeding –** The mating of animals not as closely related as the average of the population.

- There are four types of outbreeding;
  - **Species cross –** Crossing of animals of different species. (Example: Horse to donkey)
  - **Crossbreeding –** Mating of animals of different established breeds
  - **Outcrossing –** Mating of unrelated animals within the same breed
  - **Grading up –** Mating of purebred sires to commercial grade females and their female offspring for several generations

Outbreeding . . .

- **Effects of outbreeding**
- Masking the expression of deleterious recessive alleles (not elimination) through increase in heterozygosity
- Heterosis or hybrid vigor
  - Gain in favorable gene combination values
  - Increased vigor of the offspring over that of the parents
  - The more unrelated two individuals of a population, lines, or breeds are, the greater the hybrid vigor

Outbreeding . . .

- Outbreeding, usually in the form of crossbreeding, is practiced to exploit two major genetic aspects

– Hybrid vigor

– Breed complementarity: used in beef cattle, sheep, and swine

  - Requires identifying maternal and paternal breeds

    – Maternal breeds should be excellent in traits such as: fertility, freedom from dystocia, milk production, maintenance efficiency, and mothering ability

    – Paternal breeds are those strong in traits such as: rate and efficiency of gain, meat quality and carcass yield

Systems of Crossbreeding

- Crossbreeding is one of the most rapid ways of making genetic change
- Well designed crossbreeding generates hybrid vigor and breed complementarity – both very important for production efficiency
- Maintenance of acceptable levels of these two conditions in a manageable way over the long term requires a **robust crossbreeding system**
- Crossbreeding systems are the domain of commercial livestock production
- Popular way is to **use males of the new breed** by:

– Natural mating or

– AI

Systems of Crossbreeding . . .

- Crossbreeding can be used to:

– Utilize heterosis in systematic crossbreeding programs

– Upgrade local breeds

– Develop new breed from crossings of breed(s) from temperate zones with tropical breeds

- Criteria to evaluate crossbreeding systems

– Merit of component breeds each breed must bring favorable attributes to the mix
– Hybrid vigor
– Breed complementarity – the ability to produce more desirable offspring from genetically different breeds
– Consistency of performance
– Replacement considerations
– Simplicity
– Accuracy of genetic prediction

Systems of Crossbreeding . . .

- **REMEMBER:**

  – Crossbreeding is **expensive**:

- It requires more **management** planning and resources

- **Purebred herds have to be maintained** to continue crossbreeding (difficult for individual farmers)

– But… a properly designed crossbreeding:

- Allows to take advantage of breed *complementarity*

– e.g. adaptability of *Bos indicus* and production potential of *Bos taurus* cattle)

- Yields hybrid vigor or heterosis

Crossbreeding to form a new (composite/synthetic) breed

- **Purpose:** It is very **unlikely that any single breed is optimal for a specific production environment**
  - **To combine the high production potential** of exotic breeds with the adaptability of local breeds
- Formed by *inter se* mating of the first cross (F_1) or the backcross (75% exotic) for several generations
- Advantage:
  - No further outside blood is needed after the initial cross or crosses
- Disadvantage:
  - Takes long time before a new bred is developed
  - Involves risk of inbreeding
  - Decline in heterosis

Examples of composite breeds . . . Cattle

- Australian Milking Zebu (AMZ)
  - From Sahiwal, Red Sindhi, Jersey breeds

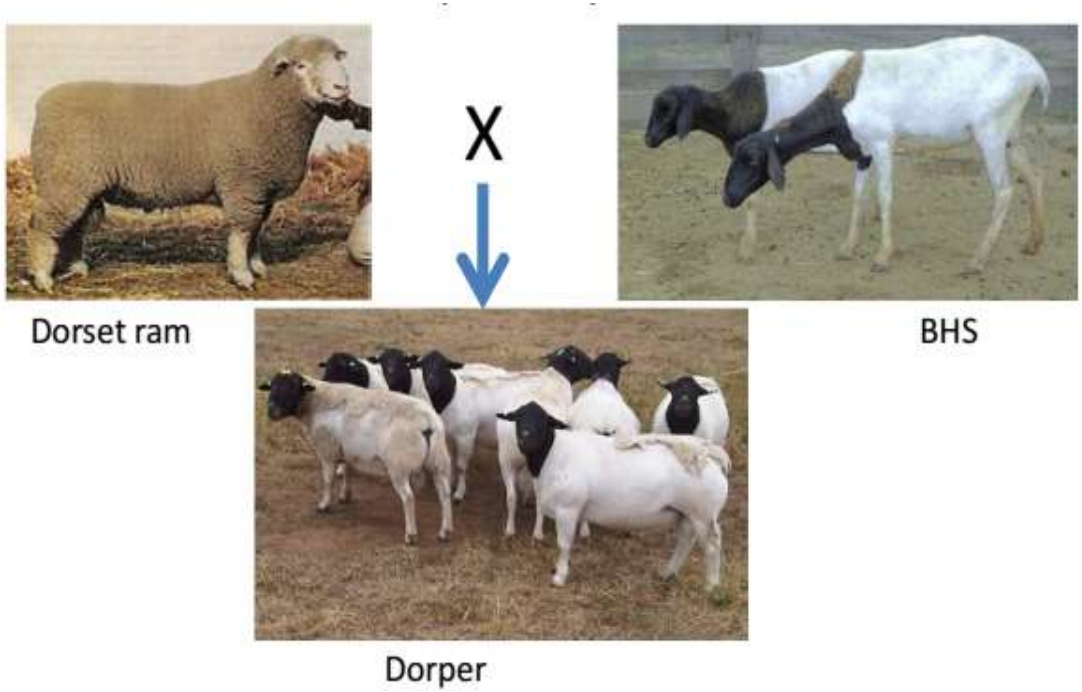


- Jamaica Hope(**Jersey-Zebu** or **Montgomery-Jersey)**
  - From Jersey, Friesian, Sahiwal breeds



Examples of composite breeds . . . sheep

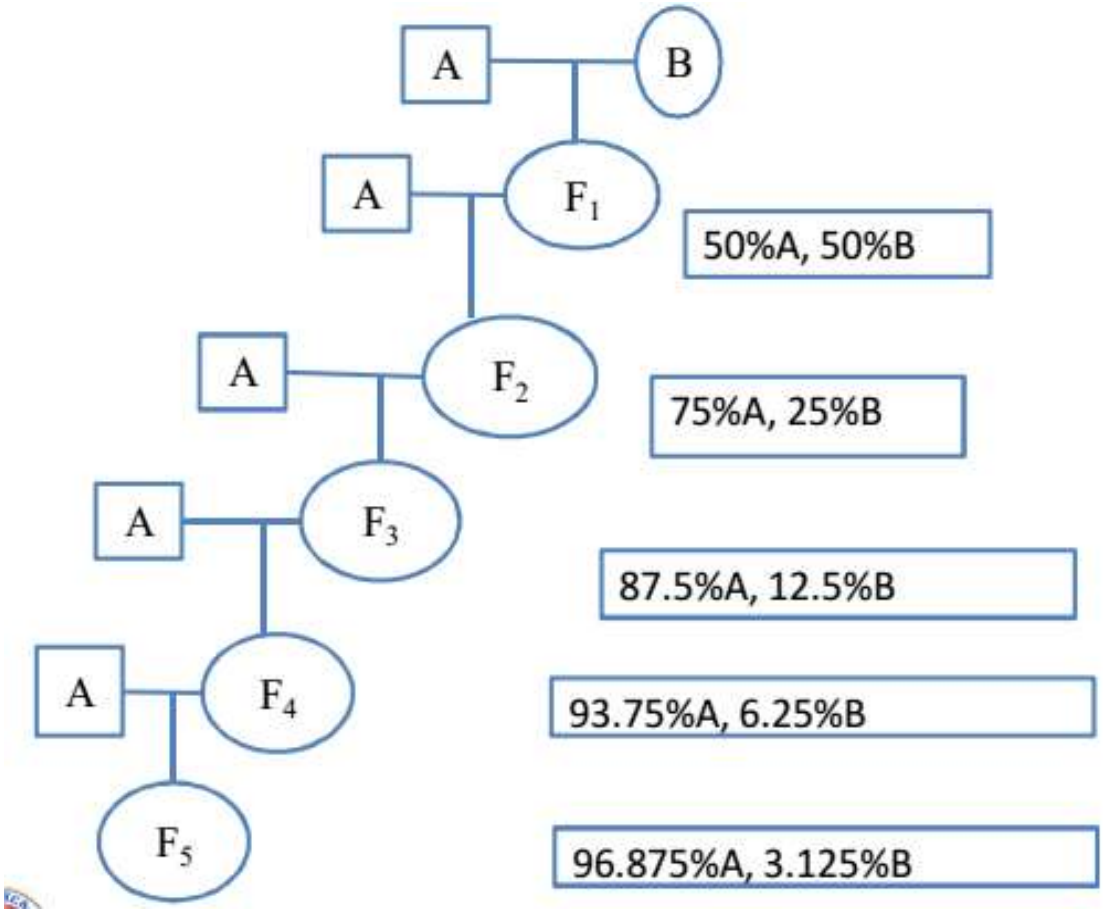
- Dorper sheep - developed from Dorset Horn and Black Head Persian (Somali) breeds



Crossbreeding to Upgrade local breeds

- Achieved through continuous backcrossing using males of one breed
- After F_5, there is little difference between genotypes of upgraded animals and animals of the desired breed
- The conditions for upgrading should not be heterosis
- **Advantage:** • Ample time to adjust the husbandry practices to fit requirements of the improved animals

Crossbreeding to Upgrade local breeds

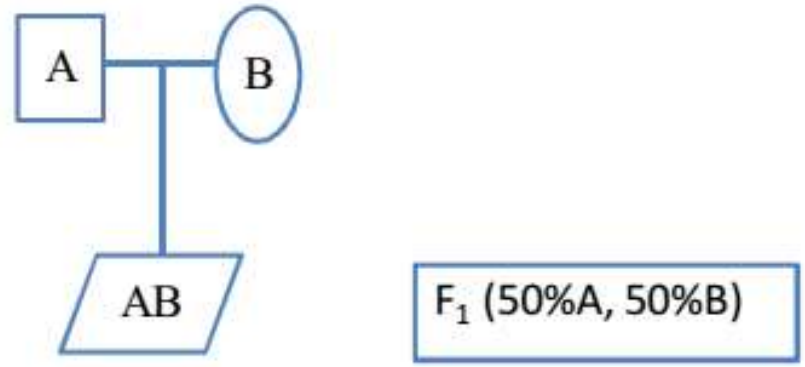


Two-way (breed) cross

- Purebred females of one breed are mated with males of a second breed to produce F1 generation which is not used for breeding

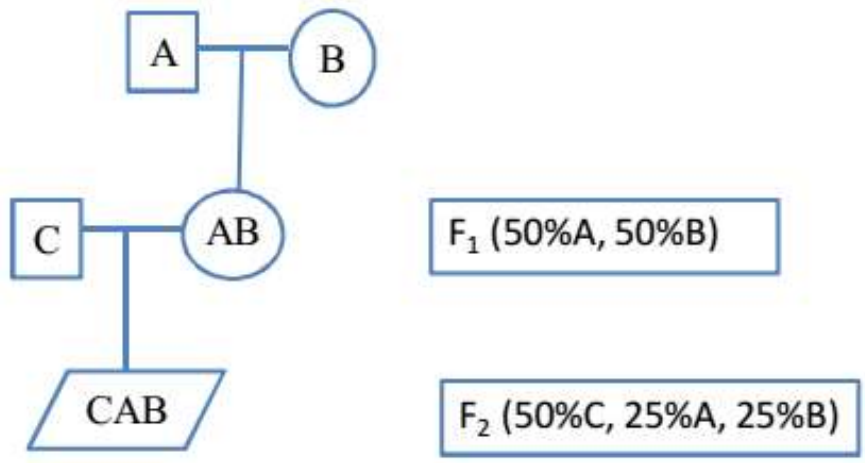
– Hybrid vigor is expressed in F1 cross

– Pure breeds must be replaced continuously



Three-way (breed) cross

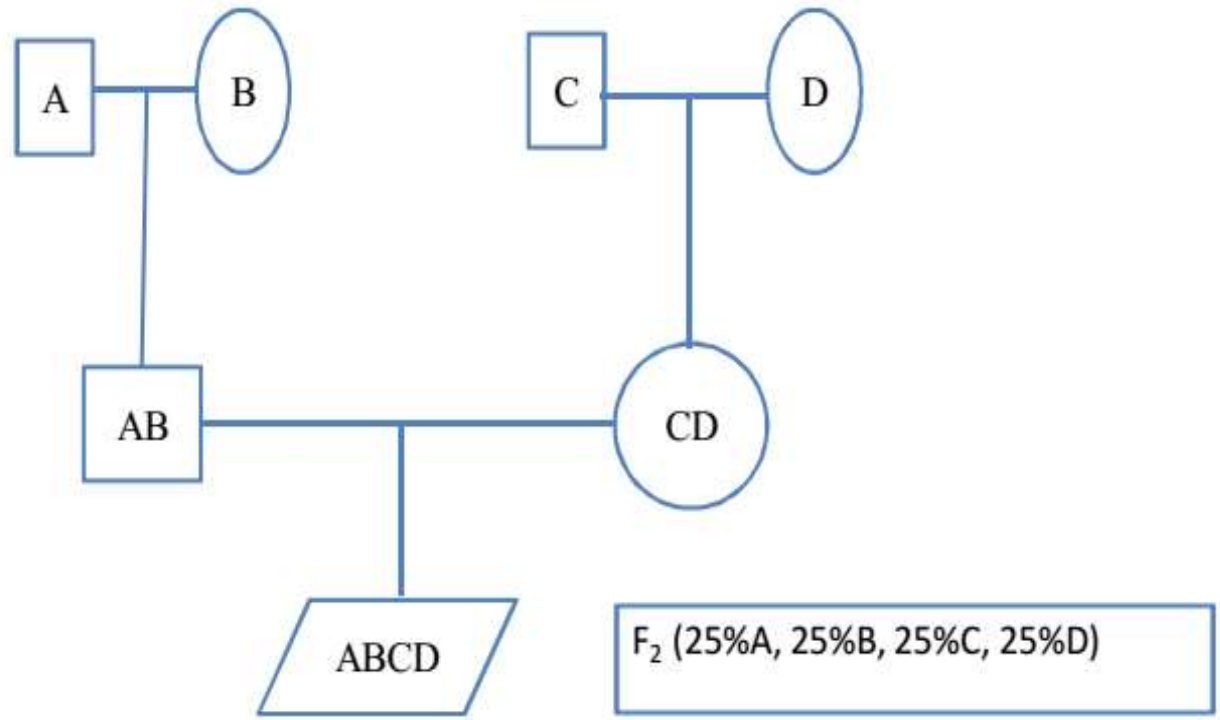
- Male animals of the third breed are mated with females of the first cross
- Utilizes hybrid vigor in reproductive parameters of the F_1 female animals (maternal and individual heterosis)



Four-way (breed) cross

- Involves four different breeds aiming to utilize paternal, maternal, and individual heterosis
  - – Common in swine and laying hen industries
    - e.g. in pigs: Duroc (breed A), Hampshire (breed B), Large White (breed C), and Landrace (breed D)
    - – A x B F boars are crossed with C x D F sows to produce terminal market animals

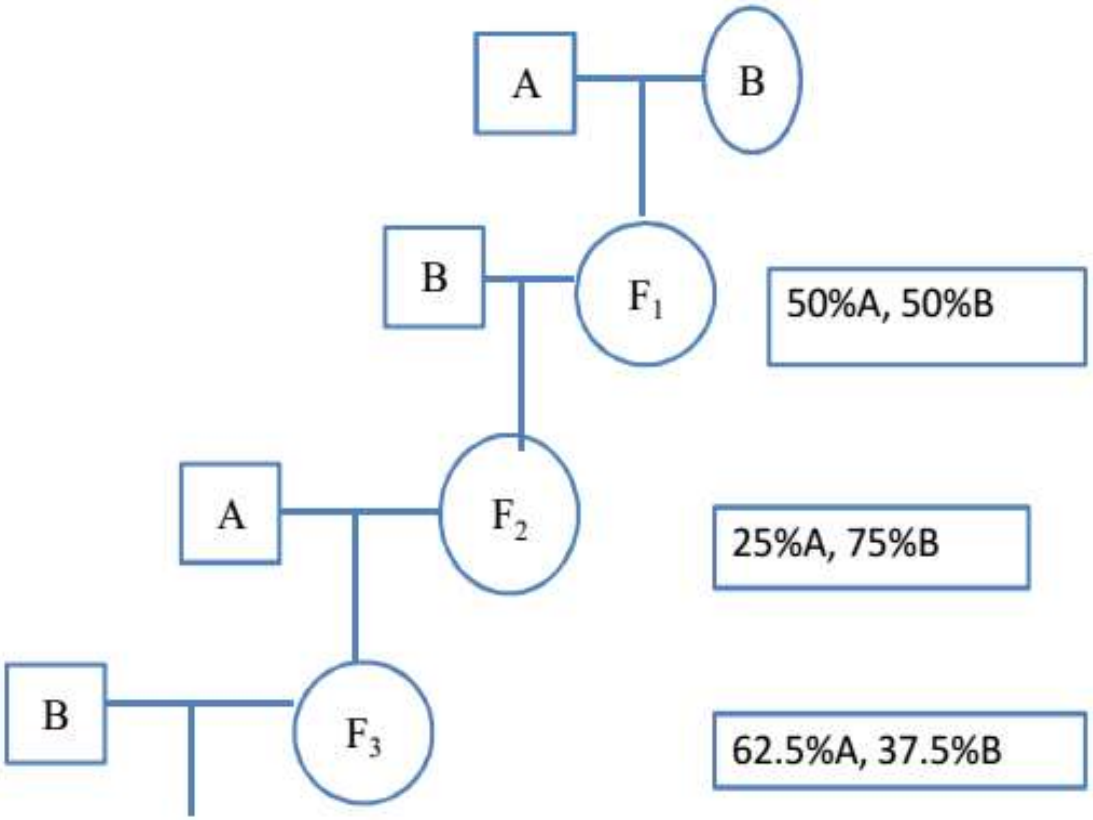
Four-way (breed) cross . . .



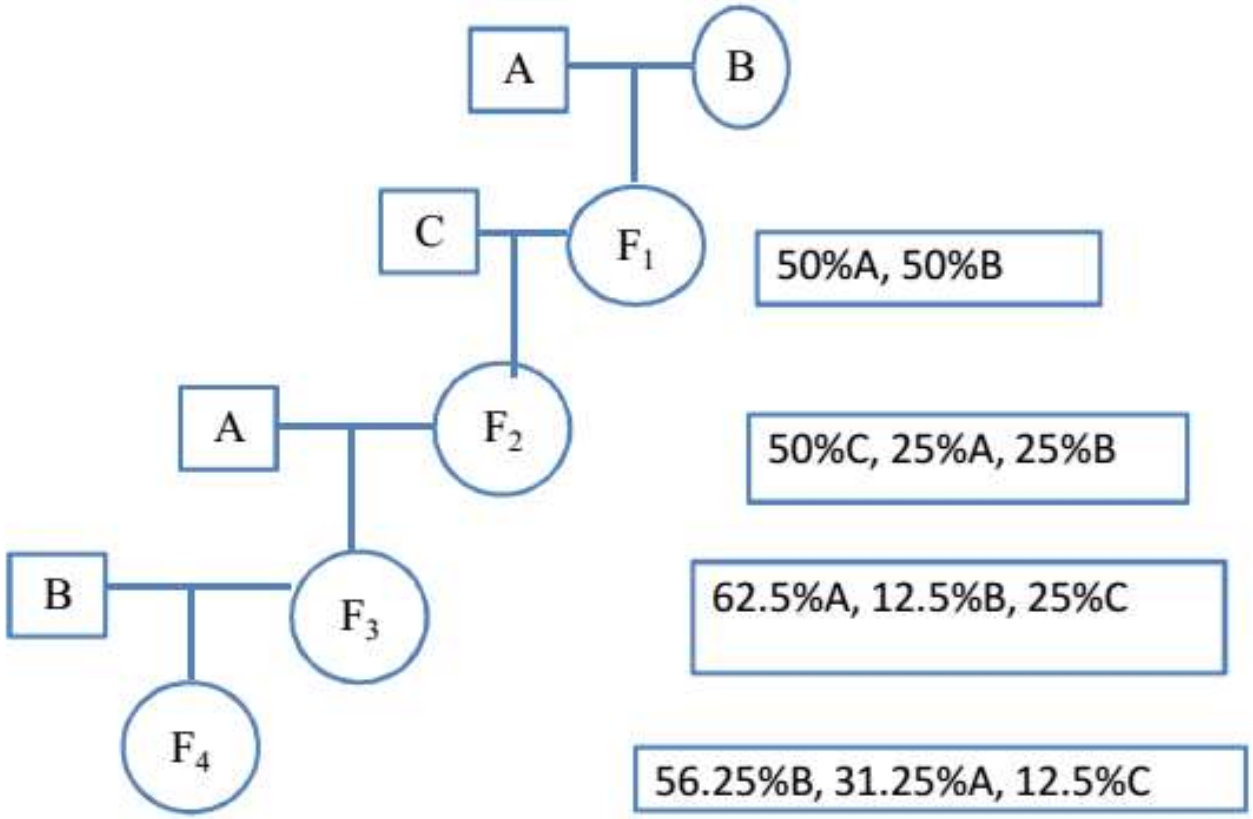
Rotational Crossing

- Can be two- or three-breed rotation
- Two-breed rotation involves backcrossing of the subsequent crossbred animals to the two purebred parental lines in turn
  - Breed composition (at equilibrium) 1/3 and 2/3
  - Unsteady performance level is seen over generations
- Three-breed rotation involves first producing an F1 female offspring to be crossed to a third breed and then the subsequent crossbreds to the three parental lines in turn
  - Breed composition (at equilibrium) 1/7, 1/7 and 4/7
  - Unsteady performance level is seen over generations

Rotational Crossing: Two-way rotational cross



Rotational Crossing: Three-way rotational cross



Exercise (rotational crossing)

- Calculate breed composition (i.e. gene proportion or blood levels of breeds) after 7 generations of rotational crossing for each of the following two conditions:
- **Two-breed rotation with breeds *A* and *B***
- **Three-breed rotation with breeds C, D and E**

- **Note:** • Show gene proportions of each breed involved in the rotational crossing system at each generation.

Heterosis

Heterosis

- Superiority of crossbred animal relative to average of its straight bred parents
  - Breed A Weaning Wt. = 530 lb.
  - Breed B Weaning Wt. = 470 lb.
  - A x B Crossbred Calf
  - Expected performance 500 lb. (average of A and B)
  - Actual performance = 520 lb.
  - 20 lb. (4%) increase = heterosis

Heterosis

- **Heterosis in Breeding**
  - Traits that are lowly heritable show high levels of heterosis, such as reproductive traits
  - Moderately heritable traits show moderate levels of heterosis, such as the growth traits
  - Highly heritable traits such as fleece and carcass traits show little hybrid vigor

Measurement of Heterosis

- In practice heterosis is measured as the difference between the average performance of the crossbreds and their purebred parent lines/breeds
- Heterosis (H)
- Heterosis with respect to gene $$\% H = \frac{Mean\ of\ F_1 - Mean\ of\ parents\ breeds}{Mean\ of\ parents\ breeds} X\ 100$$

❖ **Dominance**

• Crossing reduces the proportion of loci with homozygous recessive genes

❖ Over dominance

• The value of heterozygous genes pairs exceeds the value of homozygous genes pairs

Aa > AA, aa

❖Epistasis

• It is a type of gene interaction that occurs when an allele or **genotype at one locus 'masks' or inhibits** the expression of another allele found at different locus

Any gene that 'masks' the expression of another, non-allelic gene is epistatic to that gene.

Genes suppressed are said to be hypostatic.

Measurement of Heterosis . . .

- Estimation of H requires measurement of performance of each of the purebreds and their reciprocal cross
- Reason: reciprocal crosses may differ in performance because of maternal influence
  - e.g.: a calf from a Holstein cow bred to a Borana bull will be heavier at weaning than a calf from a Borana cow bred to a Holstein bull because the Holstein cow gives more milk than the Borana cow
- Maternal effects can be important for the offspring at the time of birth up to weaning
- Maternal effects may disappear post-weaning but may continue afterwards

Measurement of Heterosis . . .

- Example 1: consider these average weaning weights:
  - Purebred calves of Breed A = 240 kg
  - Purebred calves of Breed B = 260 kg
  - Calves from mating of A sires to B dams = 280 kg
  - Calves from reciprocal cross (B sires mated to A dams) =270 kg
  - Purebred average = (240 + 260)/2 = 250 kg
  - Crossbred average = (280 + 270)/2 = 275 kg
  - $\% H = \frac{275-250}{250} X\ 100$ = **10%**

Measurement of Heterosis . . .

- Example 2: Calculate the amount of heterosis based on the information provided ?
  - Purebred calves of breed A have an average weaning weight of 480lb, and Breed B calves have an average weaning weight of 520lb.
  - Calves resulting from the mating of Breed A sires to breed B dams have an average weaning weight of 560 lb, and calves resulting from the reciprocal cross (Breed Bsires mated to Breed A dams) have an average weaning weight of 540 lb.

Types of Heterosis

- The two most commonly exploited types of heterosis in animal breeding are individual and maternal heterosis
- **Individual heterosis**: is heterosis that results because an animal is crossbred (regardless of whether its dam is purebred or crossbred)
  - It is expressed by heavier birth weight and faster growth compared to average performance of the breeds that were crossed to produce the crossbred animal

Types of Heterosis Cont . . .

- ***Individual heterosis…***

  – Individual heterosis depends on the degree of un relatedness between the mates

  – e.g. H Levels for *Bos taurus* x *Bos taurus* crosses are low relative to levels shown for *Bos indicus* x *Bos taurus* crosses

  – Individual heterosis for weaning weight:

  - *Bos taurus* x *Bos taurus* crosses = 3.9% average
  - *Bos indicus* x *Bos taurus crosses*= 12.6%; and even greater for post weaning gain

Individual heterosis…

- However, there is also a concurrent effect of heterosis on birth weight (11.1%) which can result in substantial increase in calving difficulty when *Bos taurus* heifers are bred to *Bos indicus* bulls.
- On the other hand, calving difficulty is generally not a concern with the reciprocal cross (*Bos indicus* heifers bred to *Bos taurus* bulls) due to the inherent ability of *Bos indicus* females to restrict fetal growth

Maternal heterosis

- Expressed for traits:–
  - Measured on crossbred cows (such as calving rate, fertility)
  - Measured on their progeny (e.g. calf survival, birth weight, and weaning weight)
- Crossbred cows have higher calving rates than purebred cows

  – e.g. For *B. taurus* x *B. taurus* F1 crosses, calving rates are 3.7% higher than the purebred average, whereas they are 13.4% higher for *B. indicus* x *B.taurus* F1 crosses

Maternal heterosis

- Because fertility is such an economically important trait, even the smaller 3.7% increase due to heterosis from Bos taurus x Bos taurus crosses can have a decisive effect on profitability.
- Crossbred cows also have calves with higher survival rates and birth weights than calves of purebred cows

Maternal heterosis

- The level of maternal heterosis for calf weaning weight is particularly high, averaging 16% for Bos indicus x Bos taurus and 3.9% for Bos taurus x Bos taurus crosses
- Example: Assuming a modest average weaning weight of 475 lb for calves of purebred cows in temperate region, the calves of B. indicus x B. taurus F1 crossbred cows would be expected to weigh 16% more (or 551 lb) at weaning due to the effects of maternal heterosis alone.
- Individual heterosis effects would further increase expected weaning weight

Effect of breed type on level of heterosis (%)

	Bos taurus x Bos taurus	Bos indicus x Bos taurus
Individual heterosis		
Birth weight	2.4	11.1
Weaning weight	3.9	12.6
ADG	2.6	16.2
Maternal heterosis		
Calving rate	3.7	13.4
Calf survival	1.5	5.1
Birth weight	1.8	5.8
Weaning weight	3.9	16.0

Cundiff and Gregory, 1999